

EFY Note

The source code of this project is available for free download at source.efymag.com

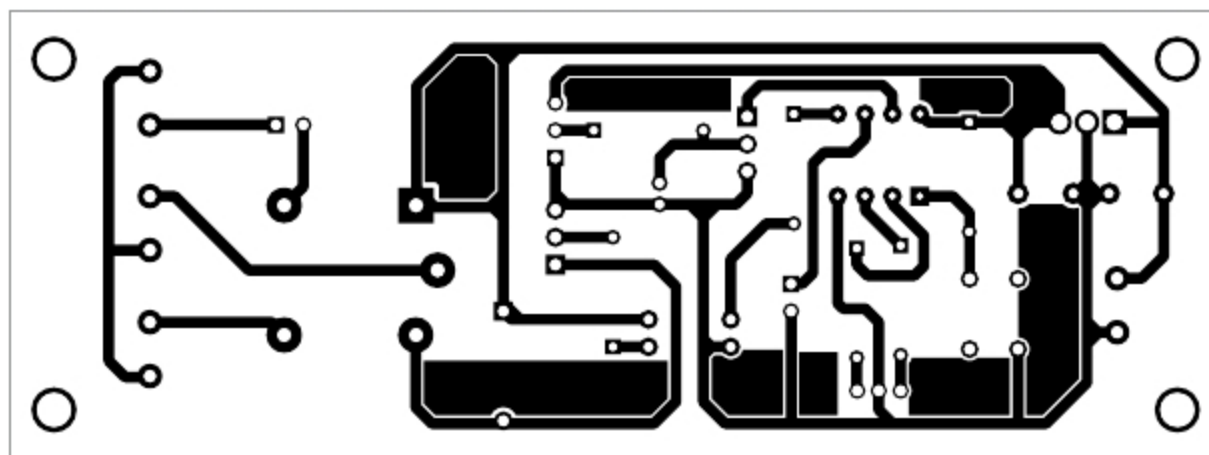


Fig. 2: Actual-size PCB layout of the smart light

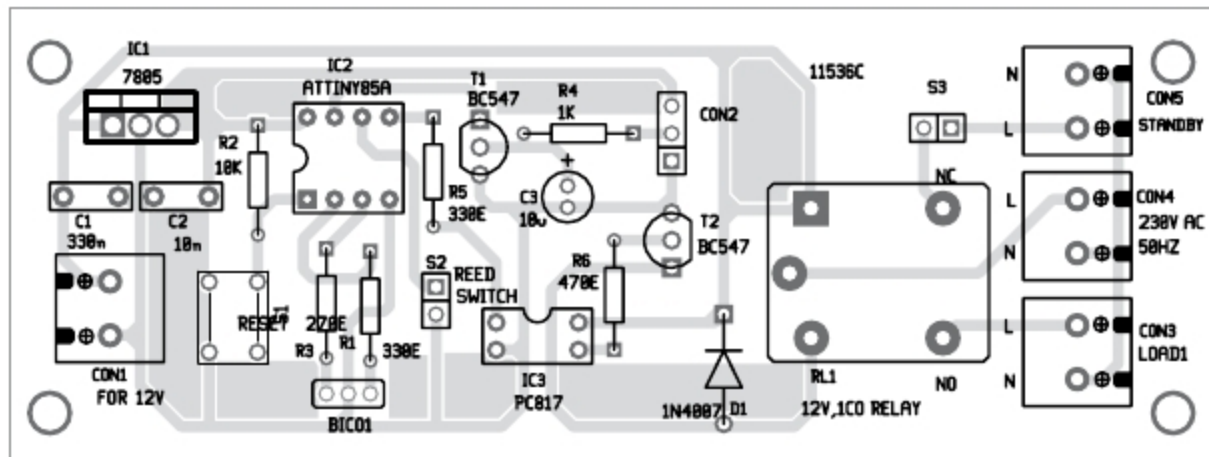


Fig. 3: Components layout for the PCB

steps given below.

1. Open File→Preferences and in Additional Boards Manager URLs enter https://raw.githubusercontent.com/damellis/attiny/ide-1.6.x-boards-manager/package_damellis_attiny_index.json.

2. Open Tools→Board→Board Manager. Scroll down the list and find ATtiny by Davis A. Mellis. Click on that and install it. Now, ATtiny will be listed in Boards menu.

3. Select ATtiny25/45/85 under Tools→Board. Select ATtiny85 under Tools→Processor. Now, select 8MHz (internal) under Tools→Clock.

Arduino as ISP. To make Arduino Uno work as ISP programmer, connect Arduino Uno to the PC/laptop and open Arduino IDE. Make sure you select Arduino/Genuino Uno as your board. Open Arduino ISP example sketch, and upload Arduino ISP sketch

to Arduino Uno. Arduino Uno is now an ISP programmer.

Programming ATtiny85A. By default, ATtiny85A runs at 1MHz. To make it run at 8MHz, select Tools→Burn Bootloader. Make sure that Arduino Uno board is programmed as ISP and it is connected with ATtiny85A through ISP pins. Now, upload the source code (trial_room.ino) to ATtiny85A using Arduino Uno board.

Construction and testing

An actual-size PCB layout of the smart light is shown in Fig. 3 and its components layout in Fig. 4. After assembling the circuit on the PCB, enclose it in a suitable box.

A small magnet bar or disk is fixed on the door and reed switch S2 is fixed on the door frame. Connect reed switch to the PCB using two wires. The reed switch and magnet

PARTS LIST

Semiconductors:

IC1	- 7805 voltage regulator
IC2	- ATtiny85A MCU
IC3	- PC817 opto-coupler
T1, T2	- BC547 npn transistor
D1	- 1N4007 rectifier diode
BICO1	- Common-cathode bi-colour LED
PIR	- PIR sensor

Resistors (all 1/4-watt, ±5% carbon):

R1, R5	- 330-ohm
R2	- 10-kilo-ohm
R3	- 270-ohm
R4	- 1-kilo-ohm
R6	- 470-ohm

Capacitors:

C1	- 330nF ceramic
C2	- 10nF ceramic
C3	- 10µF electrolytic

Miscellaneous:

CON1,	- 2-pin terminal connector
CON3-CON5	- 3-pin female Berg connector
CON2	- Tactile switch
S1	- Normally-open magnetic reed switch
S2	- On/off switch
S3	- 12V, single-changeover relay
RL1	- 230V AC-operated light bulb
LOAD1,	- 12V DC power supply (or 12V battery)
LOAD2	- PCB

(All these are available from kitsnspares.com)

must face each other, so that when the door is closed, S2 gets activated by the magnet.

Load1 is the bulb used for the smart light, while Load2 is the standby bulb. You may enable the standby low-power bulb to be continuously on till S3 is on and relay is not energised. (Relay RL1 on means Load1 is on.) **EFY**



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